REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ONE WORLD, BISHOP ALPHA AND KINDOROKO SECONDARY SCHOOLS
FIRST FORM FOUR JOINT PRE – MOCK ASSESSMENT

033/2A **BIOLOGY 2A**

(ACTUAL PRACTICAL A)

(For Both Private and School Candidates)

Time: 2:00 Hours **Year: 2025**

MARKING GUIDE

1. (a) (i) - Skin **03 Marks @1**
- Ears
- Nose

(ii) **06 Marks @1**

SENSE ORGAN	STIMULUS	RESPONSE
Skin	Touch, temperature, pain, pressure, etc.	Feeling
Ears	Sound waves	Hearing
Nose	Smell, chemicals	Smelling

(b) (i) Skin **01 Mark**

(ii) *Four non-sensory functions of the skin.* **04 Marks @1**

- *It protects the body against mechanical and chemical injuries*. Example hazardous substances like UV radiation.
- *It prevents loss of moisture*. It has the upper cover of sebum (oily secretion) which is water proof to prevent the skin from drying due to water loss.
- *It produces vitamins*. It produces vitamin D through exposure to ultraviolet radiation in sunlight.
- *It regulates the body temperature*. This is through sweat production, vasodilation and vasoconstriction
- *It provides immune function*. This is because it forms a biological barrier against pathogens. It also secretes sebum that is anti-microbial in nature.
- *It stores fats and water*. This ensures extra insulation to the body.
- *It excretes nitrogenous metabolic wastes*: The sweat secreted by the skin contains urea as metabolic wastes which are removed during sweating.

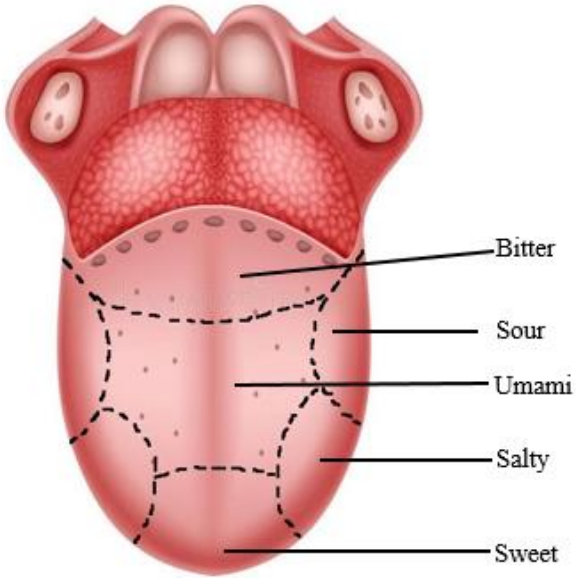
(c) (i) Tears **01 Mark**

(ii) *Two functions of tears.* **02 Marks @1**

- To wash the eyeball. This helps to remove dust particles and irritants entering the eye
- To lubricate the eyeball. This helps to reduce friction when blinking so as to aid smooth movement of the eyeball
- To fight against infection. Tears contain enzymes and antibodies that kill some microorganisms that enter the eye
- To keep the eyeball moist. This keeps the eyeball smooth and prevent dryness which may bring blurred vision
- To relieve emotions. Tears remove stress hormone (cortisol), activates the parasympathetic nervous system which relaxes the body. It also calls for attention from people which may bring comfort.

(d)

THE DIAGRAM OF THE TONGUE SHOWING POSITIONS OF THE TASTE BUDS



Caption, C = 1
Diagram, D = 1.5
Labelling, L = 2.5
Total: 05 Marks

(e) *Dangers (Problems) that a person would encounter if the sense organs stop functioning:*

Any 3 points = 03 Marks @1

- Difficult in communication. Example: Loss of hearing can lead to misunderstandings and isolation
- Occurrence of accidents and injury. Example: Without sense of touch, a person cannot feel pain from burns, cuts or extreme temperature. Also, without a sense of vision, a person may touch or step on dangerous things and cause an accident.
- Loss of balance and posture. Example: If the ears stop functioning, a person loses the sense of balance and posture leading to dizziness and frequent falls
- Difficult in performing daily duties. Everyday activities like cooking, crossing the road, identification of objects, etc. become difficult without proper sensory input.
- Nutritional risk. Example: If a person loses the sense of taste and smell, he or she can eat or drink spoiled food substances leading to poisoning.

2. (a) (i) **A** Bread mould/Mushroom **02 Marks @0.5**
B Moss plant
C Fern plant
D Hibiscus flower

(ii) **4.5 Marks @0.5**

SPECIMEN	KINGDOM	PHYLUM/DIVISION	CLASS
B	Plantae	Bryophyta	Musci
C	Plantae	Filicinophyta/Pteridophyta	Filicinae
D	Plantae	Angiospermophyta	Dicotyledoneae

(b) (i) *Three unique features of kingdom fungi*. **03 Marks @1**
- They have cell walls made up of chitin
- They feed saprophytically
- Their body is made up of mycelium that consists hyphae, except yeast

(ii) *Three advantages of the members of kingdom fungi to other organisms*. **03 Marks @1**
- *They are used as a source of food*. Some species like mushroom are edible. They are rich in proteins and vitamins especially vitamin B
- *They increase soil fertility*. Some fungi decompose organic matter into simple substances as they feed. Therefore, they add nutrients in the soil which increase soil fertility
- *They are used in brewing*. Yeast ferments carbohydrates especially glucose into alcohol. Alcohol is used as the solvent for organic substances like fats. It is also used as fuel to run engines.
- *They are used in baking*. Some species like *Aspergillus* are used in bread and cheese making in industries. Yeast, for example, is also used to raise dough for cooking bites and burns (mandazi).
- *They are used in medicines production*. Some fungi are used in the production of medicine. For example, penicillin is an antibiotic made from fungi known as *Penicillium*. It is used for treatment of diseases like tetanus.
- They are used in pest control. Fungi can be used to control pests like caterpillars of butterflies that damage the growing parts of plants. This is called biological control.
- They are used treatment of polluted water or soil. Some fungi like *Mucor*, *Aspergillus* and *Rhizopus* are used to treat polluted water or soil by decomposing organic pollutants found in the water or soil.

(c) (i) *Two reasons for why bryophytes are considered to be the most primitive plants on earth.* **01 Mark @0.5**
- They lack true roots, steams or leaves.
- They lack vascular tissues (Xylem and Phloem)

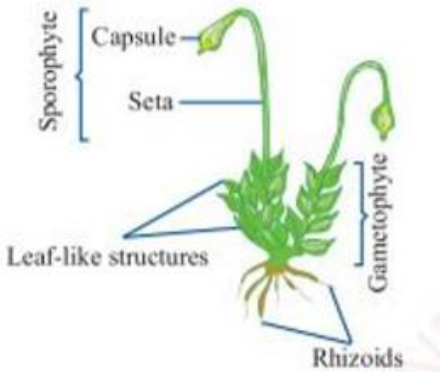
(ii) *Two similarities between specimen **B (Moss plant)** and **C (Fern plant)**:* **01 Mark @0.5**
- Both have chlorophyll (Are autotrophs/Manufacture their own food)
- Both reproduce asexually and sexually
- Both have cellulose cell walls
- Both show alternation of generation

*Two differences between specimen **B (Moss plant)** and **C (Fern plant)**:* **02 Marks @1**

SPECIMEN B (MOSS PLANT)	SPECIMEN C (FERN PLANT)
It lacks vascular tissues	It has vascular tissues
It lacks true roots, stems and leaves	It has true roots, stem and leaves
It has rhizoids	It has roots
It has capsule for spore production	It has sori for spore production
Its gametophyte generation is dominant	Its sporophyte generation is dominant
It has no fiddlehead	It has fiddlehead

(d) (i)

THE DIAGRAM OF SPECIMEN B (MOSS PLANT)



C = 0.5
D = 1.5
L = 1.5
Total: 3.5 Marks

(ii) Sporophyte generation **0.5 Mark**

(iii) - Club mosses **01 Marks @0.5**
- Horsetail

(e) (i) *Three observable adaptations of hibiscus flower for **insect** pollination*. **1.5 Marks @0.5**

- They have large brightly-coloured petals to attract insects
- They are scented to attract insects
- They produce nectar to attract insects
- They have a small sticky stigma for insects to get in contact with insects while collecting nectar.
- They have stigma and anthers held firmly in position within the flower to make them strong
- They have small anthers that produce few but large pollen grains.
- They produce large, heavy and sticky pollen grains to be picked by insects

(ii) *Two advantages of sexual reproduction*: **01 Mark @0.5**

- It leads genetic variations and hence increase adaptability of the population
- It increases the chance of survival of organisms due to their adaptability
- It reduces the risk of genetic diseases through recombination process

Two disadvantages of sexual reproduction: **01 Mark @0.5**

- It spends a lot of energy and time as it needs two parents, except in hermaphrodites
- It slows down the rate of population growth
- It can lead to mutations during gamete formation which are passed on to the offspring
- It is uncertain due to unguaranteed processes like pollination and fertilization
- It can lead to the transfer of diseases especially sexually transmitted diseases

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